

Deepfake Detectors - Zero Shot & Few shot 26-1-R-8

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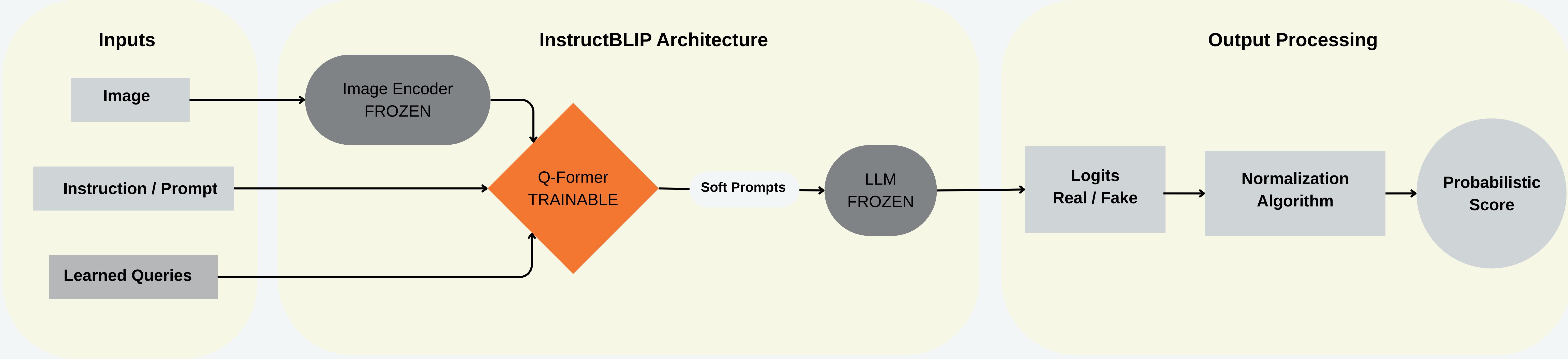


Research Problem

Deepfake technology has become widely accessible, enabling realistic synthetic media creation with minimal effort. Existing detection methods rely on low-level pixel artifacts, making them fragile against unseen attack types and prone to overfitting. This project addresses the generalization gap by leveraging Visual Language Models (VLMs) as zero-shot deepfake detectors, exploiting deep semantic understanding rather than surface-level visual cues.

Model Overview

InstructBLIP was selected for its instruction-aware Q-Former, which extracts visual features conditioned on the input prompt - enabling semantic deepfake detection without task-specific pretraining. Only the Q-Former is fine-tuned, the Vision Encoder and LLM remain frozen.



Visual Datasets

CelebA-HQ (SimSwap)

41,712 balanced images (20,856 real / 20,856 fake) generated via SimSwap. Used exclusively for Zero-Shot & cross-dataset evaluation.

DFDC (Kaggle Subset)

Pre-extracted facial frames from the Deepfake Detection Challenge with diverse real-world conditions. 112,183 images were Used for fine-tuning & cross-dataset evaluation.

Celeb-DF-v2

159,968 frames sampled from high-quality celebrity deepfake videos (32 per video). Used for fine-tuning & cross-dataset evaluation.

Probabilistic Deepfake Scoring Algorithm

Since VLMs generate free text, we intercept the model at the first token and extract raw logits for all "Yes"/"No" variants. Taking the MAX logit per class and applying Softmax over the pair yields a calibrated P(Fake) score - enabling continuous AUC evaluation without an external classifier.

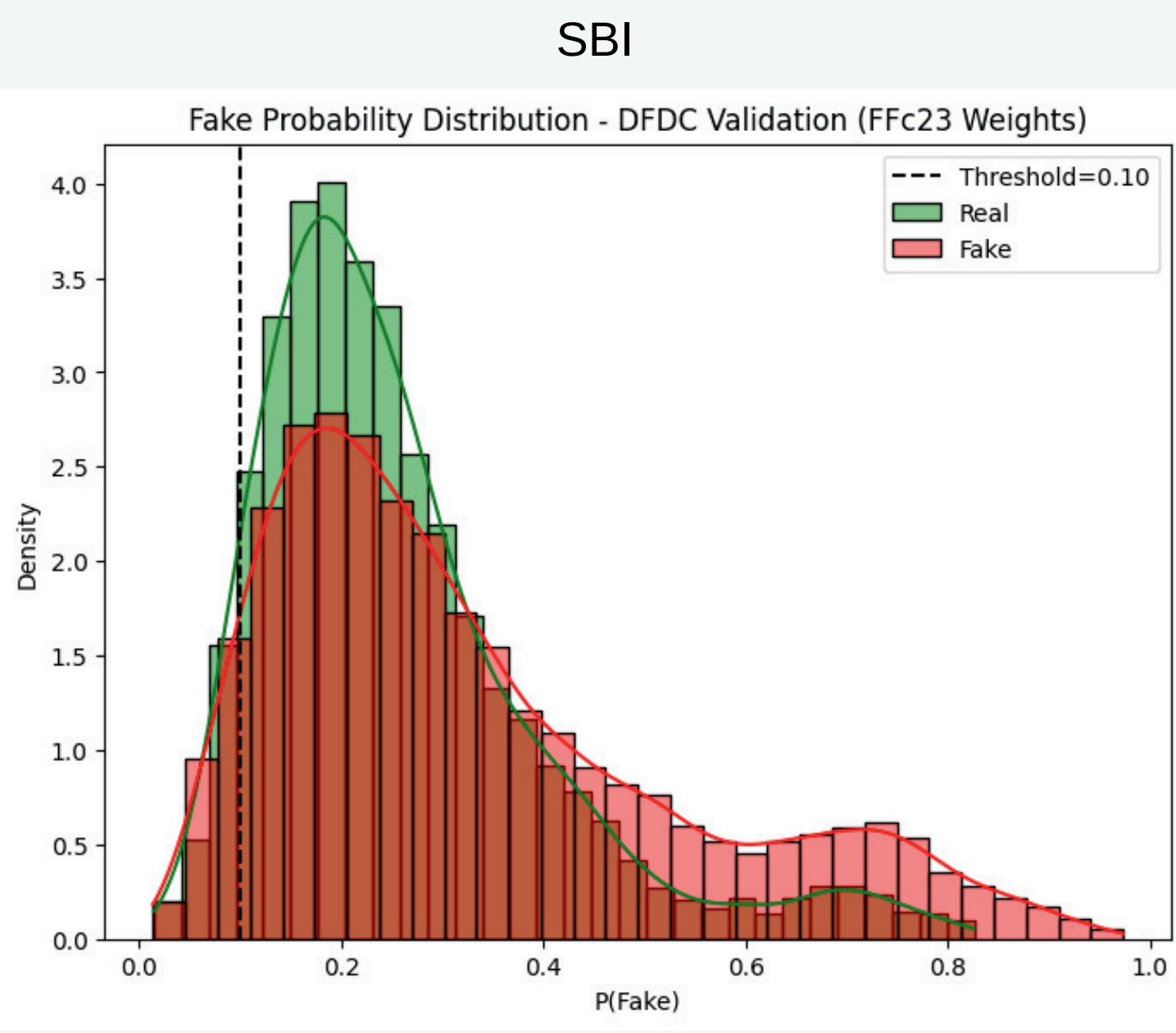
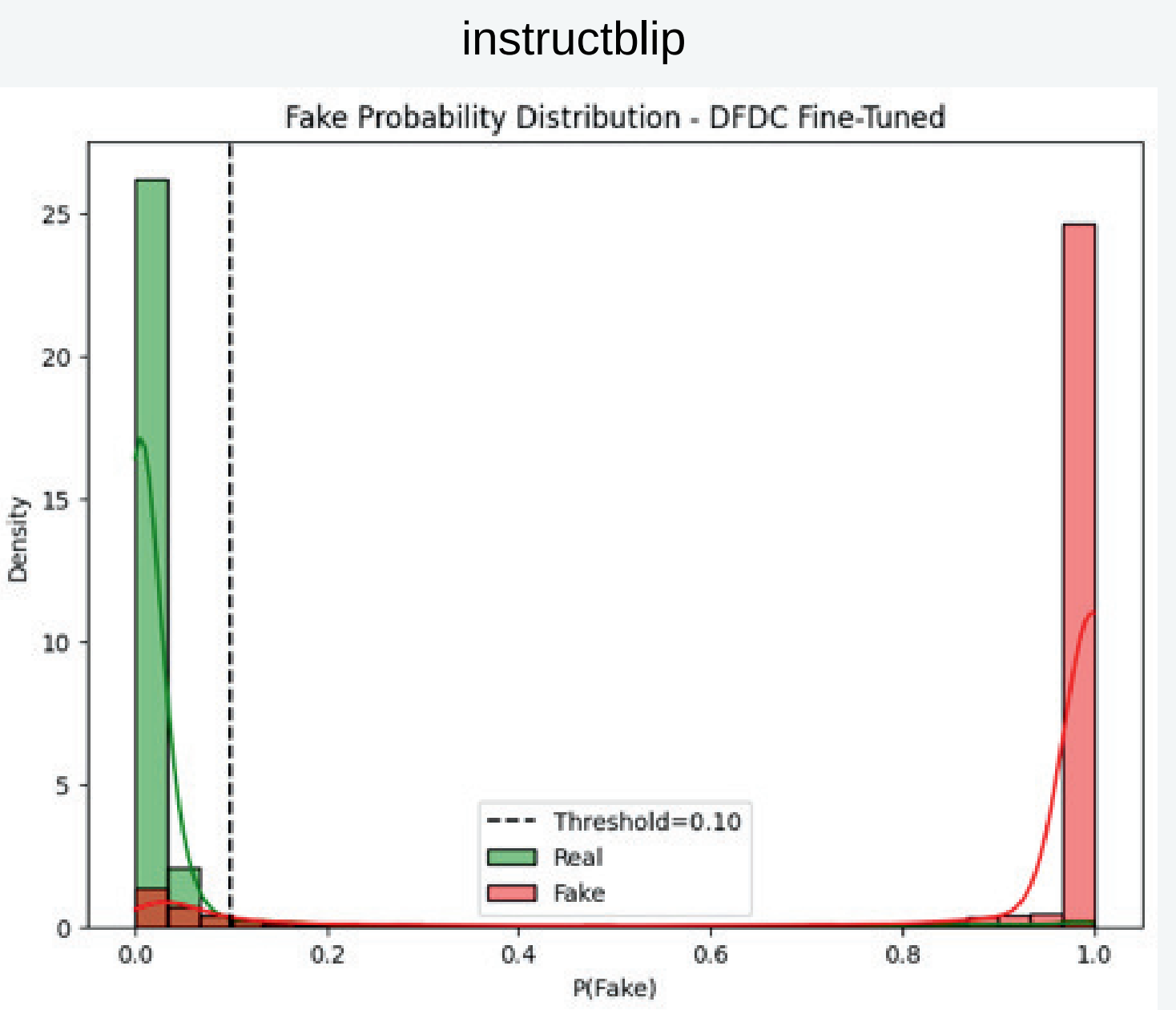
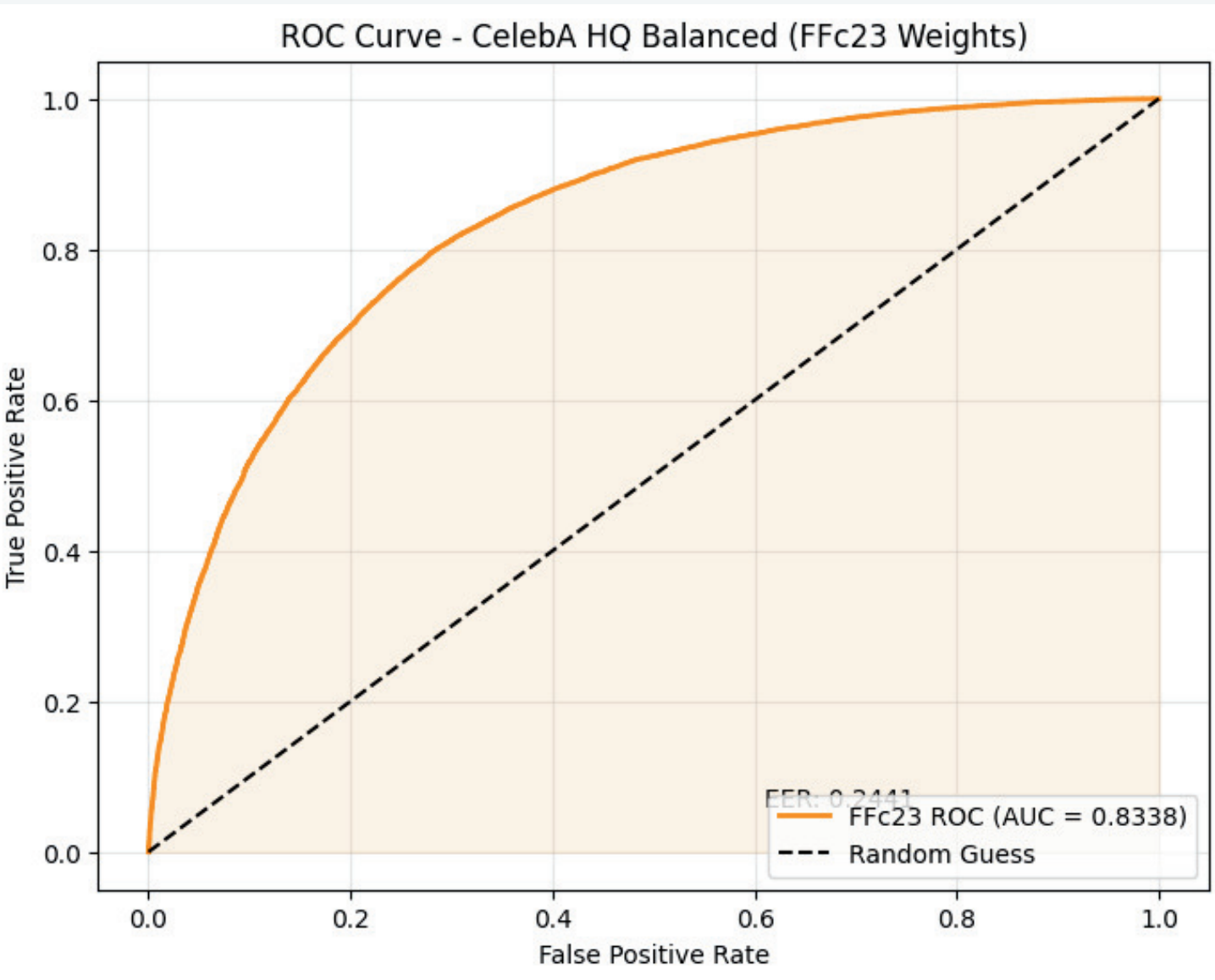
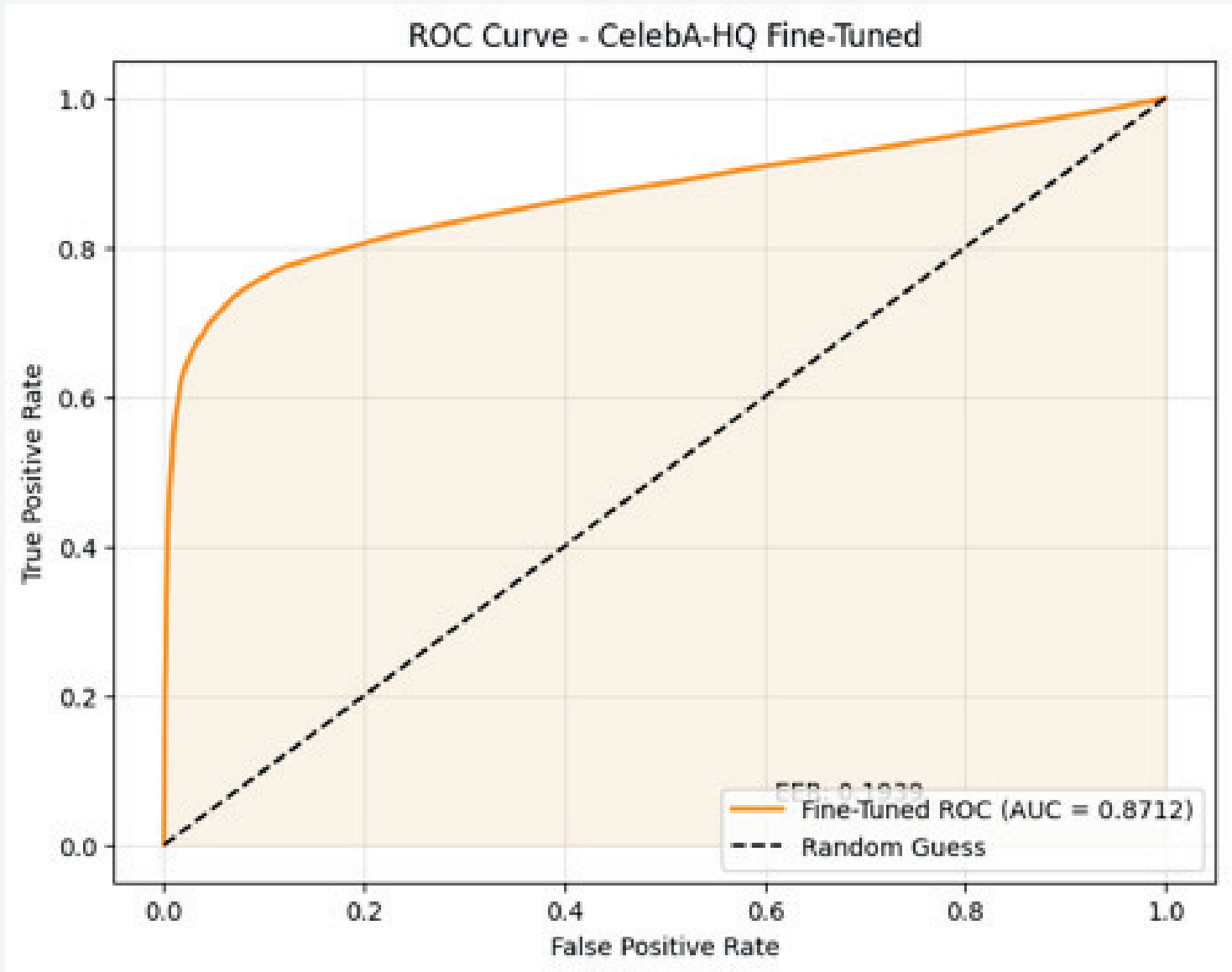
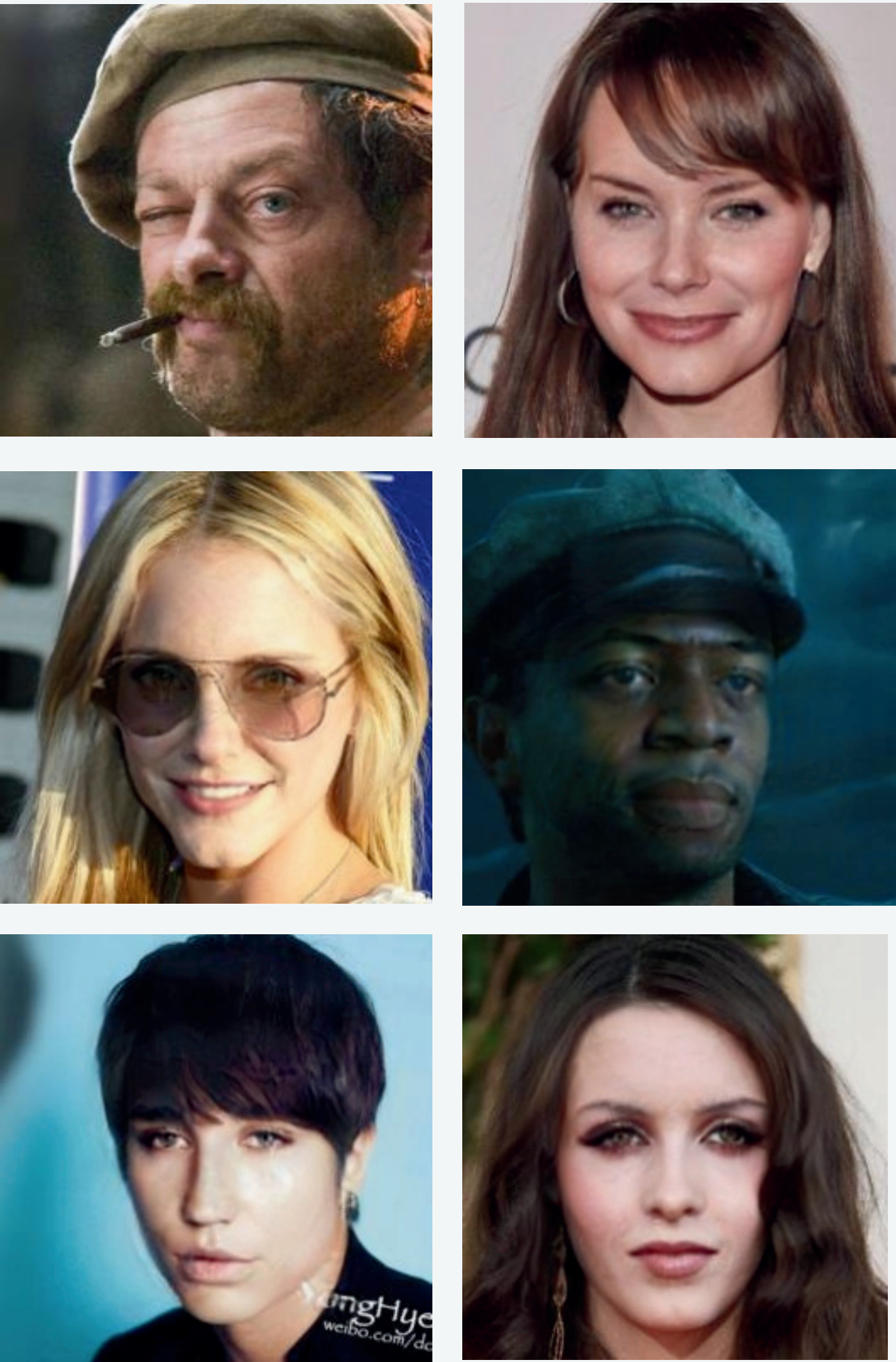
the Algorithm:

Image + Prompt → Frozen Vision Encoder & Q-Former → single-token generation → extract logits for {Yes, No} variants → MAX per class → Softmax → P(Fake)

Results And Graphs

Results compare Zero-Shot and Few-Shot InstructBLIP against SBI and Pirogov (2025), evaluated via ROC-AUC. Fine-tuning on DFDC (LR=5e-5) achieved AUC=0.9808 in-domain and AUC=0.8712 cross-dataset, outperforming Zero-Shot (0.7406) and surpassing SBI (0.8338) on this unseen manipulation type - though SBI (0.93) still edges out our Celeb-DF-v2 tuned model on its home domain. ROC curves confirm strong discrimination; probability distributions show clear Real/Fake separation post fine-tuning. Embedding-space analysis (t-SNE, Silhouette Score 0.05-0.21) shows classification is driven by the model's terminal decision layer rather than linearly separable features.

Try It Yourself:
Can You Spot The Fake?



Configuration	Dataset	AUC
Pirogov (2025) - Zero-Shot	CelebA-HQ	0.813
Ours - Zero-Shot	CelebA-HQ	0.7406
Pirogov (2025) - Fine-Tuned	DFDC	~0.98
Ours - Fine-Tuned (LR=5e-5)	DFDC	0.9808
Ours - Fine-Tuned (LR=5e-5)	CelebA-HQ (cross)	0.8712
SBI Pipeline- FFc23	CelebA-HQ	0.8338
SBI Pipeline- FFc23	DFDC	0.5949

